



UNIVERSITAS
INDONESIA

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FAKULTAS

ILMU
KOMPUTER

PENGANTAR STATISTIKA

Intro to Probability

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Learning Outcomes

After this lesson, students should be able to:

1 Model uncertainty using events

Define experiments, sample spaces, and events; represent relationships using complement, union, intersection, and subsets.

2 Apply fundamental probability rules

Use probability axioms, basic properties, addition and complement rules, and equally likely outcomes to solve problems.

3 Reason with conditional probability

Compute and interpret conditional probabilities and use the Law of Total Probability to combine mutually exclusive cases.

4 Update probabilities using Bayes' theorem

Use prior information and new evidence to calculate posterior probabilities in Information Systems contexts.

5 Assess independence between events

Determine whether events are independent and apply the multiplication rule appropriately.

Outline

01

Probability Foundations

Experiment • Sample space • Events • Long-run frequency

02

Algebra of Events

Complement • Union • Intersection • Mutually exclusive events • Subsets • De Morgan's laws

03

Basic Probability Rules

Axioms • Basic properties • Addition rule • Complement rule • Equally likely outcomes

04

Conditional Probability

Conditional probability • Multiplication rule • Law of Total Probability

05

Bayes' Theorem

Prior • Likelihood • Evidence • Posterior • Worked applications

06

Independence

Independent events • Testing independence • Multiplication rule for independent events

Why Probability Matters

Probability gives us a language for uncertainty.

Observe

A sample is obtained from a population.
We usually assume an underlying probability distribution.

Describe

Use descriptive statistics to summarize what the sample tells us.

Decide

Use inferential statistics to make decisions about the problem.

Probability theory provides the foundation for statistical inference from data.

01 Probability Foundations

Build the language for describing uncertainty.

Experiment → Sample Space → Event → Probability



Example: login attempt → {success, fail} → “login fails” → P(fail)

Sample Space

Experiment

Any process or procedure for which more than one outcome is possible.

Sample Space (S)

The set of all possible outcomes of an experiment.

Coin toss

$S = \{\text{Head, Tail}\}$

Six-sided die

$S = \{1, 2, 3, 4, 5, 6\}$

Login attempt

$S = \{\text{Success, Fail}\}$

The sample space must describe every possible outcome relevant to the experiment.

Events

An event E is a subset of the sample space S .

Dice

$S = \{1, 2, 3, 4, 5, 6\}$

$E = \text{"even score"} = \{2, 4, 6\}$

System status

$S = \{\text{Up}, \text{Degraded}, \text{Down}\}$

$E = \text{"service unavailable"} = \{\text{Down}\}$

**If the observed outcome
belongs to E ,
we say that event E occurred.**

Complement of an Event

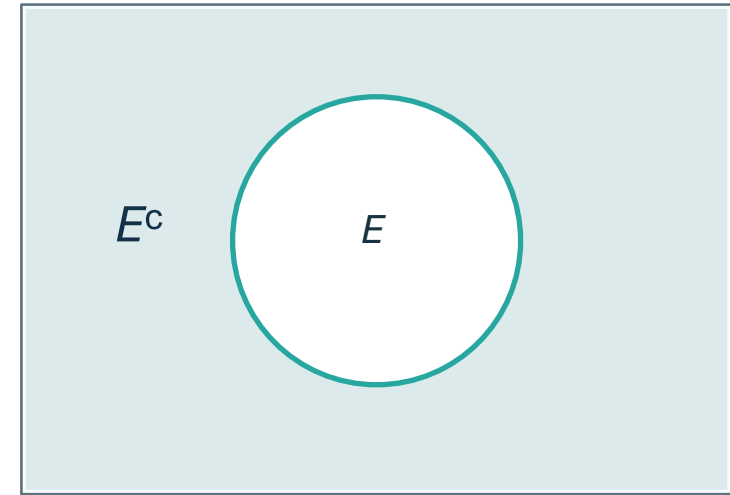
E^c contains all outcomes in S that are not in E .

Example

$S = \{1, 2, 3, 4, 5, 6\}$

$E = \{1, 3, 5\}$

$E^c = \{2, 4, 6\}$



$$P(E^c) = 1 - P(E)$$

Think First: What is S^c ?

Union of Events

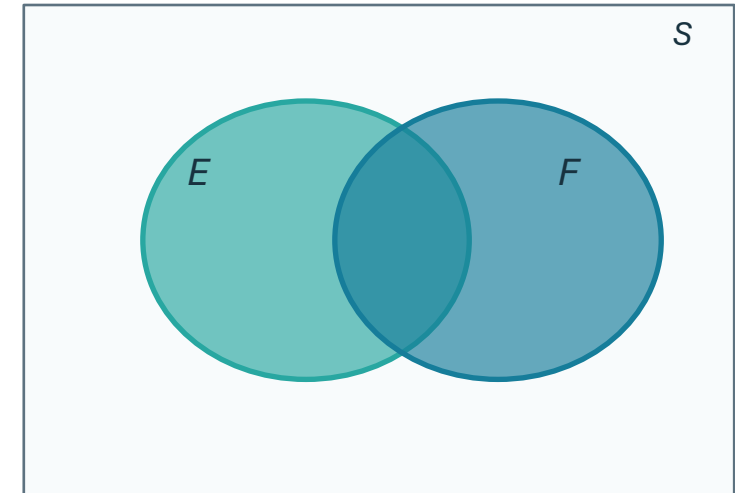
$E \cup F$ contains outcomes that are in E , in F , or in both.

Example

$$E = \{1, 3, 5\}$$

$$F = \{2, 4, 6\}$$

$$E \cup F = \{1, 2, 3, 4, 5, 6\} = S$$



Union means OR

Intersection & Mutually Exclusive Events

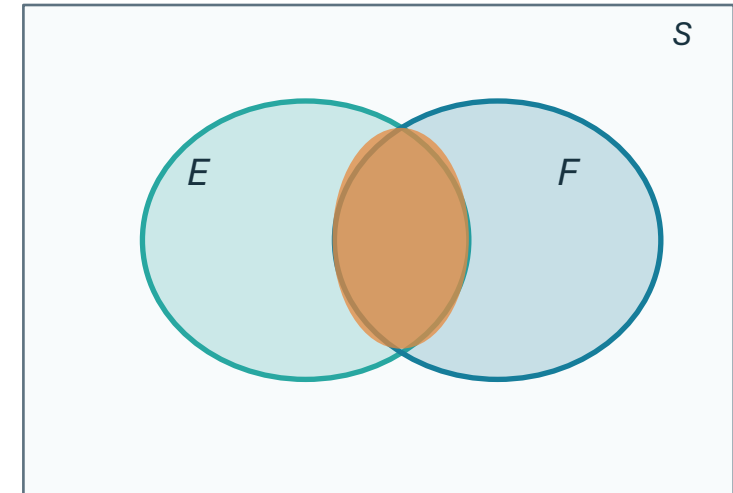
$E \cap F$ contains outcomes that are in both E and F .

Information Systems example

E = “transaction > Rp1M”

F = “transaction flagged as suspicious”

$E \cap F$ = transactions satisfying both conditions



If $E \cap F = \emptyset$, then E and F are mutually exclusive.

Null event: $\emptyset$

Subset & Proper Subset

Subset ($\subseteq$)

$$\{a,b\} \subseteq \{a,b,c\}$$

$$\{a,b,c\} \subseteq \{a,b,c\}$$

Proper subset ($\subset$)

$$\{a,b\} \subset \{a,b,c\}$$

$$\{a,b,c\} \subset \{a,b,c\} \text{ X}$$

If $E \subseteq F$ and $F \subseteq E$, then $E = F$

02 Algebra of Events

Describe how events relate to one another.

Algebra of Events

Commutative

$$E \cup F = F \cup E$$
$$E \cap F = F \cap E$$

Associative

$$(E \cup F) \cup G = E \cup (F \cup G)$$
$$(E \cap F) \cap G = E \cap (F \cap G)$$

Distributive

$$(E \cup F) \cap G = (E \cap G) \cup (F \cap G)$$
$$(E \cap F) \cup G = (E \cup G) \cap (F \cup G)$$

These rules let us manipulate events just like algebraic expressions.

De Morgan's Laws

$$(E \cup F)^c = E^c \cap F^c$$

$$(E \cap F)^c = E^c \cup F^c$$

Probability: How Likely Is an Event?

Question 1

Will the server fail tonight?

Question 2

Will a user complete checkout?

Question 3

Will the KRL be disrupted today?

Question 4

Will a transaction be fraudulent?

Probability turns uncertain questions into quantities we can reason about.

Probability & Long-Run Frequency

Repeat the same experiment under the same conditions.

10 trials

0.60

100 trials

0.53

1,000 trials

0.501

10,000 trials

0.500

As repetitions increase, the proportion of times E occurs approaches a constant value:
 $P(E)$.

03 Basic Probability Rules

Turn event relationships into probability calculations.

Axioms of Probability

Axiom 1

For every event E:
 $0 \leq P(E) \leq 1$

Axiom 2

For the sample space S:
 $P(S) = 1$

Axiom 3

For mutually exclusive events:
 $P(\cup E_i) = \sum P(E_i)$

$P(E)$ is the probability assigned to event E.

DERIVED FROM THE AXIOMS

Basic Probability Properties

$$P(\emptyset) = 0$$

$$P(E^c) = 1 - P(E)$$

$$\text{If } A \subseteq B, \text{ then } P(A) \leq P(B)$$

$$P(E \cup F) = P(E) + P(F) - P(E \cap F)$$

Why subtract $P(E \cap F)$? Because the overlap was counted twice.

Worked Example: Software Defects

Event A

A = “no more than two errors”
= {0 errors, 1 error, 2 errors}

$P(0)=0.05$ $P(1)=0.08$ $P(2)=0.35$

$$P(A) = 0.05 + 0.08 + 0.35 = 0.48$$

$$P(A^c) = 1 - 0.48 = 0.52$$

Interpretation: the probability of more than two errors is 0.52.

Worked Example: At Least One Course

A student has a probability of $\frac{2}{3}$ of passing Mathematics and a probability of $\frac{4}{9}$ of passing Biology. The probability that the student passes both Mathematics and Biology is $\frac{1}{4}$. What is the probability that the student passes at least one of the two courses?

$$P(M \cup B) = P(M) + P(B) - P(M \cap B)$$

$$= \frac{2}{3} + \frac{4}{9} - \frac{1}{4} = \frac{31}{36}$$

“At least one” → think UNION.

Practice: Events and Complements

A web service request has outcomes {Success, Slow, Timeout, Error} with probabilities 0.1, 0.3, 0.5, 0.1.

A

{Success, Slow}

B

{Slow, Timeout, Error}

C

{Error}

Find:

- ① $P(A)$, $P(B)$, $P(C)$
- ② $P(A^c)$, $P(B^c)$, $P(C^c)$
- ③ $P(A \cap B)$, $P(A \cup B)$, $P(A \cap C)$

Think First: Is $P(\text{rolling a 6}) = 1/6$?

A die is rolled. What is the probability that its face equals 6?

$1/6$?

Not automatically.

The answer $1/6$ is correct only if all six outcomes are assumed **equally likely**.

Equally Likely Outcomes

Suppose $S = \{1, 2, \dots, N\}$ and every outcome has the same probability p .

$$1 = P(S) = Np \Rightarrow p = 1/N$$

$$P(E) = \text{Number of outcomes in } E / \text{Number of outcomes in } S$$

This counting rule requires the equally likely assumption.

Fair Dice: From One Die to Two

One fair die

$S = \{1,2,3,4,5,6\}$
 $E = \text{“even”} = \{2,4,6\}$
 $P(E) = 3/6 = 1/2$

Two fair dice

36 ordered outcomes
Each outcome has probability $1/36$

(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

Highlighted: sum = 6 → 5 outcomes → $P = 5/36$

Two Dice: Both Scores Are Even

A: first die is even B: second die is even

(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

$P(A \cap B) = 9/36 = 1/4$

INTERSECTION = both
conditions must be true.

Two Dice: At Least One Even Score

A: first die is even B: second die is even

(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

$$P(A \cup B) = 27/36 = 3/4$$

UNION = at least one condition is true.

04 Conditional Probability

Update the relevant sample space when information is known.

Conditional Probability

The probability of event E given that event F has occurred is called conditional probability.

$$P(E | F) = P(E \cap F) / P(F), \quad \text{for } P(F) > 0$$

INTUITION

Once F is known to occur, F becomes the relevant sample space.

CONDITIONING EVENT

F is called the conditioning event.

Conditional Probability: Useful Identities

$$P(E \mid F) + P(E^c \mid F) = 1$$

$$\text{If } E \cap F = \emptyset, \text{ then } P(E \mid F) = 0$$

$$\text{If } F \subseteq E, \text{ then } P(E \mid F) = 1$$

KEY IDEA

Within F , E and E^c still partition all possibilities.

Conditioning changes which outcomes are relevant.

Example: Red Die = 5, Given Sum = 6

A red die and a blue die are thrown.

E: red die = 5

F: sum of both dice = 6

$$P(E | F) = (1/36) / (5/36) = 1/5 = 0.20$$

		Die 2					
Die 1	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)	
	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)	
	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)	
	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)	
	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)	
	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)	

Among the five outcomes in F, only (5,1) also satisfies E.

Example: Successful Login Given Correct Password

A university information system records login attempts. Out of 100 login attempts, 80 users enter the correct password. Among those 80 attempts with the correct password, 72 result in a successful login. Given that the password is correct, what is the probability that the login is successful?

Once we know that the password is correct, we only consider the 80 attempts with a correct password.

NEW SAMPLE SPACE

80 login attempts
with correct password

FAVORABLE OUTCOMES

72 successful
login attempts

CONDITIONAL PROBABILITY

successful among
correct-password attempts

$$P(\text{Success} \mid \text{Correct Password}) = 72 / 80 = 0.90$$

Answer: Given that the password is correct, the probability of a successful login is 90%.

Conditioning by Cases

An event E can be split according to whether another event F occurs or not.

$$E = (E \cap F) \cup (E \cap F^c)$$

Because the two parts are mutually exclusive:

$$P(E) = P(E \mid F)P(F) + P(E \mid F^c)P(F^c)$$

Compute an overall probability by combining separate cases.

Example: Accident Probability

Thirty percent of policyholders are classified as accident-prone. The probability of having an accident is 0.40 for an accident-prone policyholder and 0.20 for a non-accident-prone policyholder. If a policyholder is selected at random, what is the probability that they will have an accident?

ACCIDENT-PRONE

$$P(F)=0.30$$

$$P(A|F)=0.40$$

NOT ACCIDENT-PRONE

$$P(F^c)=0.70$$

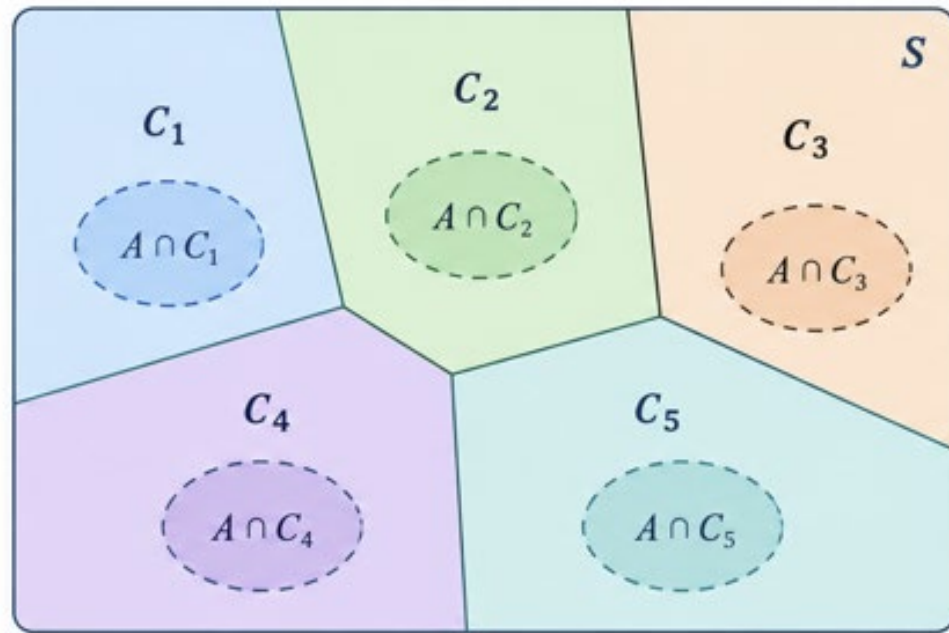
$$P(A|F^c)=0.20$$

$$P(A) = (0.40)(0.30) + (0.20)(0.70) = 0.26$$

Overall accident probability = 26%.

Law of Total Probability

Let $C_1, C_2, \dots, C_n$ be mutually exclusive events that together cover the sample space S .



 = part of event A within each case C_i

Law of Total Probability

$$P(A) = \sum_i P(A | C_i) P(C_i)$$

- 1 Partition S into mutually exclusive events $C_1, C_2, \dots, C_n$.
- 2 Compute $P(A | C_i)$ in each case.
- 3 Weight each result by $P(C_i)$, then add them.

$P(A)$ is a weighted average of the conditional probabilities.

Law of Total Probability: IT Support Ticket Escalation

An organization receives IT support tickets through four channels. Every ticket comes from exactly one channel: Web Portal (40%), Mobile App (30%), Email (20%), or Chatbot (10%). The probability that a ticket requires escalation to a specialist is 0.02, 0.05, 0.08, and 0.10, respectively. What is the probability that a randomly selected support ticket requires escalation?

C₁ • Web Portal

$$P(C_1)=0.40$$

$$P(A | C_1)=0.02$$

C₂ • Mobile App

$$P(C_2)=0.30$$

$$P(A | C_2)=0.05$$

C₃ • Email

$$P(C_3)=0.20$$

$$P(A | C_3)=0.08$$

C₄ • Chatbot

$$P(C_4)=0.10$$

$$P(A | C_4)=0.10$$

A = ticket requires escalation

$$P(A) = \sum_i P(A | C_i)P(C_i)$$

$$= (0.02)(0.40) + (0.05)(0.30) + (0.08)(0.20) + (0.10)(0.10)$$

$$= 0.008 + 0.015 + 0.016 + 0.010$$

$$P(A) = 0.049 = 4.9\%$$

Interpretation: A randomly selected IT support ticket has a 4.9% probability of requiring escalation to a specialist.

05

Bayes' Theorem

Update beliefs after observing new evidence.

Bayes' Theorem

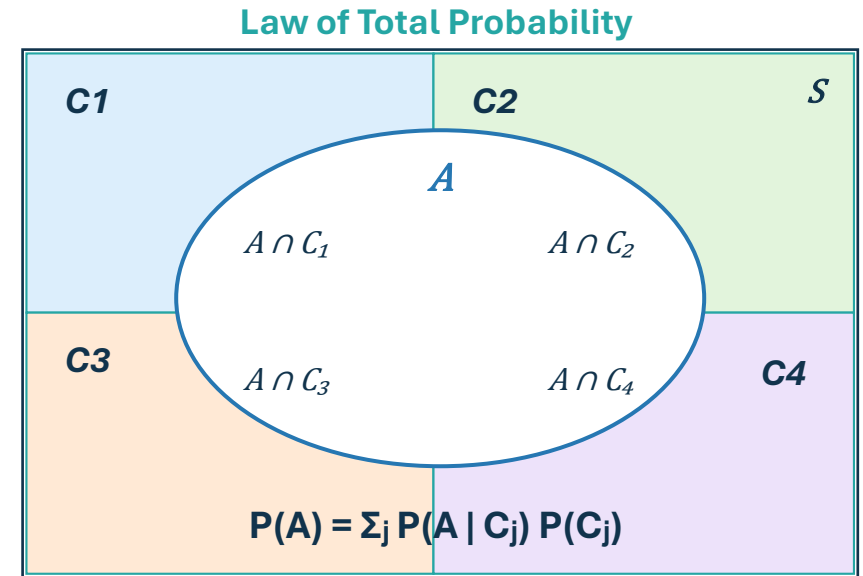
Suppose we know ...

$P(C_1), P(C_2), \dots, P(C_n)$ Prior probabilities

$P(A | C_1), P(A | C_2), \dots, P(A | C_n)$ Likelihoods

We want to compute ...

$P(C_1 | A), P(C_2 | A), \dots, P(C_n | A)$ Posterior probabilities



$$P(C_i | A) = P(C_i \cap A) / P(A)$$

$$= \frac{P(A | C_i) P(C_i)}{P(A | C_1)P(C_1) + P(A | C_2)P(C_2) + \dots + P(A | C_n)P(C_n)}$$

Bayes' theorem updates a prior probability using observed evidence A.

Worked Example : Did the Student Know the Answer?

A student is answering a multiple-choice question. The probability that the student knows the correct answer is 0.40. If the student knows the answer, the response is correct. If the student does not know the answer and guesses, the probability of answering correctly is 0.20. The student gives a correct answer. What is the probability that the student actually knew the answer?

KNOWS THE ANSWER

$$P(K) = 0.40$$
$$P(C | K) = 1$$

DOES NOT KNOW / GUESSES

$$P(K^c) = 0.60$$
$$P(C | K^c) = 0.20$$

QUESTION

Given a correct answer C , find $P(K | C)$.

Solution

$$P(C) = P(C | K)P(K) + P(C | K^c)P(K^c)$$
$$= (1)(0.40) + (0.20)(0.60) = 0.52$$

$$P(K | C) = P(C | K)P(K) / P(C)$$
$$= (1)(0.40) / 0.52 \approx 0.769$$

Answer: Given that the response is correct, the probability that the student knew the answer is approximately 76.9%.

Worked Example: Intrusion Detection

An Intrusion Detection System (IDS) monitors network traffic for cyberattacks. The probability that an attack occurs is 0.05. When an attack is present, the IDS generates an alert with probability 0.99. When no attack is present, the IDS still generates an alert with probability 0.10.

(a) FALSE ALARM

What is the probability that the IDS generates an alert when no attack is present?

(b) MISSED DETECTION

What is the probability that an attack is present but the IDS does not generate an alert?

DEFINE THE EVENTS

A: an attack is present
B: the IDS generates an alert

Try to express each requested probability using A, B, and their complements before calculating.

Intrusion Detection: False Alarm & Missed Detection

GIVEN

$P(A)=0.05$
 $P(A^c)=0.95$
 $P(B|A)=0.99$
 $P(B|A^c)=0.10$

COMPLEMENT

$P(B^c|A)$
 $= 1 - P(B|A)$
 $= 1 - 0.99 = 0.01$

RULE

$P(E \cap F)$
 $= P(E)P(F|E)$

(a) False Alarm

No attack, but an alert is generated:

$$P(A^c \cap B) = P(A^c)P(B | A^c) = (0.95)(0.10) = 0.095 = 9.5\%$$

(b) Missed Detection

Attack present, but no alert is generated:

$$P(A \cap B^c) = P(A)P(B^c | A) = (0.05)(0.01) = 0.0005 = 0.05\%$$

Interpretation: false alarms are much more frequent than missed detections under these assumptions.

Intrusion Detection: False Alarm & Missed Detection

A security alert is evidence. Bayes asks: after seeing the alert, how likely is an attack?

$$P(A | B) = P(A)P(B | A) / [P(A)P(B | A) + P(A^c)P(B | A^c)]$$

$$= (0.05)(0.99) / [(0.05)(0.99) + (0.95)(0.10)] \approx 0.3426$$



Even a strong detector can have many false alerts when attacks are rare.

06 Independence

Recognize when knowing one event does not change another.

Independent Events

Events E and F are independent if knowing that one occurred does not change the probability of the other.

$$P(E \cap F) = P(E) P(F)$$

EQUIVALENT, IF $P(F) > 0$

$$P(E | F) = P(E)$$

EQUIVALENT, IF $P(E) > 0$

$$P(F | E) = P(F)$$

Independent $\neq$ mutually exclusive.

Independence: Two Different Intuitions

PHYSICALLY SEPARATE PROCESSES

Toss a coin, then roll a die.

E: die is even

T: coin is tail

$P(E \cap T) = P(E)P(T) = 0.25$
→ independent.

SAME PHYSICAL PROCESS

Roll one die.

E: outcome even → $3/6$

F: outcome ≤ 4 → $4/6$

$E \cap F = \{2, 4\} \rightarrow 2/6$

$(3/6)(4/6) = 2/6 \rightarrow$ independent.

Independence is a probability relationship—not necessarily separate physical processes.

Two Dice: Pairwise Checks

Two fair dice are thrown.

E_7 : sum of two dice = 7

F: first die = 4

T: second die = 3

Die 1

Die 2

(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)

E_7 and F

Independent

E_7 and T

Independent

E_7 and $(F \cap T)$

Not independent

More Than Two Events

For three events E, F, and G to be independent, all pairwise products and the three-way product must hold.

$$P(E \cap F) = P(E)P(F)$$

$$P(E \cap G) = P(E)P(G)$$

$$P(F \cap G) = P(F)P(G)$$

$$P(E \cap F \cap G) = P(E)P(F)P(G)$$

For n events, the product condition must hold for every subset.

Pairwise Independent $\neq$ Mutually Independent

Perform two independent coin tosses.

A

Head on toss 1

B

Head on toss 2

C

Both tosses are equal

A, B, and C are pairwise independent.

But $P(A \cap B \cap C) \neq P(A)P(B)P(C)$

Pairwise independence alone is not enough for mutual independence.

Key Takeaways

1

Conditional probability asks: how likely is E after F is known?

2

Total probability combines cases weighted by their prior probabilities.

3

Bayes reverses the condition: from $P(\text{evidence} \mid \text{cause})$ to $P(\text{cause} \mid \text{evidence})$.

4

For SI decisions, always consider base rates, not only model accuracy.